

Problem Set 1

ECON 2010 – Macroeconomics

Spring 2018

Due at the start of class Week 3, January 31. No late problem sets will be accepted. Please refer to the syllabus section on late assignments.

Name:

How problem sets will work in this class.

Problem sets (i.e. homework like this) will work slightly differently than what I imagine you are used to from other classes. Although you will work on your problem set independently outside of class (you may not work with other students, ask for help from anyone else, or look at resources beyond your textbook and any notes and handouts provided in class), we will spend a good deal of time in class going over the correct approaches to solving the problems and the correct answers. This will help you better see where your understanding of the material is incomplete, and how to correctly go about solving the problems.

Going over the problems in class will take the following format. First, you will work in small groups assigned during class to share and compare your answers to a particular problem, or part of a problem. By doing so, you'll see alternative strategies that your classmates may have used to solve the problems, and explain to others why you got the answer that you did. In your small group, you will together arrive at a *group* answer to the problem. If you've all come up with the same answer on your own, this may be easy. If you've come up with different answers, you'll have to discuss amongst yourselves to figure out who correctly solved the problem (or even, perhaps, come up with a *new* solution if you realize that no one solved it correctly).

Finally, each group will share their group answer with the rest of the class. When you share your answer, you'll explain why your group decided on that answer, and you'll field any questions that the rest of the class has about the solution. When you see other answers shared, you'll have a chance to ask your classmates and your instructor any questions that you may have about the problem. Together, as a class, we'll confirm the correct answer to each problem and the correct solution strategy. **If the correct answer discussed in class is the same as your answer, you'll simply write "correct" under that problem on this sheet. If the correct answer discussed in class is different from your answer, you'll need to write on this sheet the correct answer and, critically, why it differs from yours.**

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.

2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
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Problem 1 (27 points)

In many large urban areas discussions of rent control have recently become prominent. Rent control is a government policy that controls the price of rental housing — specifically, rent control is a cap on prices (a price ceiling) that prevents landlords from charging rents above a certain level or increasing rents at a rate higher than a pre-determined percentage each year. Those who favor rent control hope that by implementing the policy they can maintain affordable housing for those with lower incomes.

Nashville is one city with an affordable housing crisis that has been considering a rent control policy. The median rent for a one bedroom apartment in Nashville in 2017 was \$1,397 and has been increasing rapidly over the previous decade. We can use economic thinking to help us examine some of the possible consequences for Nashville were it to adopt a rent control policy. Specifically, we can think about how the rent control policy affects the price of rental units, the demand for rental units, their quantity of rental units supplied, and even the quality of rental units.

- (a) There are many different landlords, renters, apartment buildings, apartment units, apartment sizes, rents, etc. in Nashville. This might make it appear difficult to consider the consequences of rent control. Give an example of a simplifying assumption that an economist might use to consider the effects of rent control in Nashville and explain briefly – in just a few sentences – why it might be helpful to make that assumption. **(4 points) Remember: Don't write on this problem set. See the problem set instructions.**

- (b) Rent control policy is not the only thing that might affect rents and the quantity of housing in Nashville, nor is it the only thing that might affect how many apartment units are available for rent. Give an example of another variable (it could be a policy or some other factor) that an economist might want to hold fixed in order to understand the consequences of rent control in Nashville all else equal (*ceteris parabis*). Explain briefly – in just a few sentences – why this variable might affect rents and/or the quantity of housing in Nashville and therefore why it is a good variable to hold fixed. **(4 points)**
- (c) Consider Marta, a Nashville landlord who owns a 10 unit apartment building. Before the rent control policy, she charged \$1,250 per month to rent each of her units. The rent control policy affects the price of housing and limits Marta to charging only \$1,100 per unit for each of her units per month. What is the *annual (yearly)* difference in *total* rental income that Marta receives from her *building* under rent control versus under no rent control? **(5 points)**
- (d) Now consider Riza, another Nashville landlord. Riza owns a 6 unit apartment building. Before the rent control policy, Riza rented each unit in her building for \$975. She had considered renovating each of the apartments with new windows, a new refrigerator, and a brand new stove. The cost of these upgrades is a one-time expense of \$1,300 per unit. If the rent control policy limits Riza to charging only \$1,100 per unit per month, will Riza make back the cost of the renovations this year? That is, does she have an *incentive* to undertake the renovations to improve the quality of her units (if we think only in terms of this year)? If yes, how much more than the cost of the renovations will she make? If no, how much will she lose renovating?

(5 points)

- (e) Consider the property development company, Prism LLC, that is considering building a new luxury 50-unit high-rise apartment complex in Nashville. Because Prism is building a new building, the Nashville rent control policy only requires the building to reserve 10% of all of the units as affordable housing, with rental rates capped at \$1,100 for those units. If Prism can recoup the cost of building the complex within 20 years, they'll supply these new units to the market by building the complex; if not, they won't. If it will cost Prism \$2 million to build the new apartment building, and the most renters are willing to pay for the non-rent controlled apartments is \$2,500 per month, will Prism recoup the initial expense of building the complex within 20 years? (Assume that \$1 today is worth the same as \$1 20 years from now; that is, don't worry about real versus nominal distinctions.) **(7 points)**

- (f) Consider the following excerpt from an article on rent control in Nashville. Select one sentence that represents a *positive* statement about rent control, and one sentence that represents a *normative* statement. **(2 points)**

“Estimates that as many as 100 people a day are moving to Music City mean that the supply of affordable and workforce housing is disappearing. Property owners can raise rents to

charge market rates or use the dwellings as short-term rentals, which benefit visitors but not residents. Properties leased through home sharing websites like Home-Away and Airbnb would not be subject to the rent control policy under discussion. The State Assembly should act quickly to amend the legislation to prevent landlords from removing their rental properties from the regular market and listing them with home sharing sites. Closing this loophole will prevent loss of rental supply."

Problem 2 (16 points)

Consider a very simple economy that produces only two goods — potatoes and beef. The following table represents the amount of potatoes that can be produced for different amounts of beef produced. Alternatively, you can think of the table as representing amount of beef that can be produced for different amounts of potatoes produced.

Table 1: Production of potatoes and beef

Quantity of potatoes, millions of lbs	Quantity of beef, millions of lbs
0	150
300	0
250	40
100	125
210	80

- (a) Use the table above to draw (graph) a production possibilities frontier (PPF) for this simple potatoes and beef economy on an x-y plane. The x – axis on your graph should be the quantity of potatoes and the y – axis should be the quantity of beef. Label your axes. Do your best to draw your production possibilities frontier to scale, but of course it doesn't have to be

perfect. **(5 points)**

(b) Given the production possibilities frontier, for each of the following points, is the production of potatoes and beef at this point *possible*? Note that you may be better off using calculations for some of these rather than just looking at your graph. **(3 points total, 1 point each)**

(1) The point where the quantity of potatoes is 90, and the quantity of beef is 90.

(2) The point where the quantity of potatoes is 260, and the quantity of beef is 75.

(3) The point where the quantity of potatoes is 400, and the quantity of beef is 140.

(c) Given the production possibilities frontier, for each of the following points, is the production of potatoes and beef at this point *efficient*? Note that you may be better off using calculations

for some of these rather than just looking at your graph. **(3 points total, 1 point each)**

(1) The point where the quantity of potatoes is 90, and the quantity of beef is 90.

(2) The point where the quantity of potatoes is 260, and the quantity of beef is 75.

(3) The point where the quantity of potatoes is 400, and the quantity of beef is 140.

- (d) Assume that there is only one input into the production of potatoes and beef: land. That is, producers in the economy decide whether to use land to grow potatoes or raise cattle for beef, and nothing else is needed (no labor, no equipment, etc.). (Notice that this is a simplifying assumption!) Draw a *new* production possibilities frontier to reflect an economy in which when 300 million of lbs of potatoes are produced, no beef is produced, *and* all land is *equally* as productive for producing potatoes as beef. Label your x and y axes the same as your previous plot, and label the point where the PPF crosses the x-axis. At this crossing point, how much beef is produced when no potatoes are produced? **(5 points)**

Problem 3 (20 points)

The history of the market for airline flights in the United States is an interesting one that not many people know about. Before 1978, the airline industry was heavily regulated by the federal government. The government dictated to each airline company which routes it could fly and how much it could charge for a ticket on that route.

Because airlines could not compete on price for the same route, they had to find other ways to get passengers to choose their airline over another. One way that they did this was by offering lots of flights at lots of times and offering flexible tickets to passengers. Even if your ticket was for a particular flight at a particular time, the flexible ticket meant that whenever you showed up to the airport, you would be able to get on the next plane to your destination.

In order to guarantee that passengers could get on a flight whenever they showed up at the airport, airlines had to run many more flights than they had passengers. (Otherwise you might show up at the airport to find the next planes were full! You might look for flights on another airline in the future as a result.) This resulted in a lot of nearly empty airlines flying around.

After 1978, the airline industry was de-regulated. The government stopped dictating which airlines could fly which routes and stopped setting prices. This is the (relatively) free airfare market that we're familiar with today.

- (a) Consider the period under regulation when airline companies were flying many nearly-empty planes. It cost TWA, an airline company, \$50,000 per day in fixed costs to pay airport fees, leases on its airplanes, the reservation system, and advertising. TWA had to pay these fixed costs regardless of how many flights it operated. In addition, it cost \$20,000 to operate its 10 daily flights. Let's consider what would happen if TWA wanted to add another daily flight to, say, better serve customers who wanted an early morning flight option. What would the marginal cost have been to TWA of adding this 11th daily flight? **(5 points)**

- (b) Consider the same information from part (a) above. If TWA can charged \$100 per ticket on this 11th flight, would it have been sensible to run this flight with only 25 passengers? Explain briefly why. **(5 points)**

- (c) Again, consider the same information from part (a) above. If TWA could have again charged \$100 per ticket on its 11th flight, would it have been sensible to run this flight with only 15

passengers? Explain briefly why. **(5 points)**

- (d) Before deregulation, an airline ticket from New York to San Francisco cost \$400 in 1978 dollars. This price was set by the government. In 2017, the average price for a ticket on that same route was \$480. Has deregulation caused airline fares to increase? Explain in a few sentences. **(5 points)**

Problem 4 (31 points)

Consider an early human living a few thousand years ago. This person — let's call him Bob — needed three things to survive — food, clothing and shelter. In this simple early economy, Bob lived in isolation and had to provide all three life necessities for himself. Consider food in the form of fish that Bob had to catch, clothing in the form of fabric that Bob had to weave himself (and sewed into clothes), and shelter in the form of wood that Bob cut down from trees and chopped up (and built into a dwelling).

- (a) In an 8 hour day, Bob on his own could catch 36 fish. (They're small fish.) In two hours, Bob could weave 3 feet of cloth. And in a morning of 4 hours (or afternoon) could cut down and chop up 12 trees. If Bob chooses to spend an hour cutting down and chopping trees, how many trees can he cut down and what is his opportunity cost of doing so in terms of fish caught and cloth woven? **(4 points)**.

(b) In Bob's production of food, clothing and shelter in part (a), we've effectively assumed that Bob exhibits *constant marginal returns* to his labor. That is, the first hour he spends fishing is *just as productive* as his last hour spent fishing, just as his first hour of weaving is as productive as his last (or any other), and the same for chopping trees. Come up with your own new numbers for how many trees Bob can chop in the morning versus the afternoon to illustrate Bob's productivity if he exhibited *diminishing marginal returns* to his labor. **(4 points)**

(c) Now consider that Bob doesn't live in quite so much isolation. Instead, there's another person who has to provide herself with the same three life necessities who lives on the other side of the hill from Bob. Sarah is a better fisherwoman than Bob. In a day, she can catch 44 fish. (Remember, they're small!) But she's a worse weaver and a worse tree cutter. In two hours, Sarah can only weave 2 feet of cloth, and in a 4 hour morning (or afternoon), she can only cut down and chop 6 trees. If Sarah chooses to spend an hour cutting down and chopping trees, how many trees can she cut down, and what is her opportunity cost of doing so in terms of fish caught and cloth woven? **(4 points)**

(d) Given the opportunity costs for Bob that you found in part (a) and for Sarah that you found

in part (c), answer the following. **(6 points, 2 points each)**

(1) Who has a higher opportunity cost of chopping trees as compared to catching fish?
That is, which person has to give up catching more fish to chop down a tree?

(2) Who has a higher opportunity cost of chopping trees as compared to weaving cloth?
That is, which person has to give up weaving more cloth to chop down a tree?

(2) Who has a higher opportunity cost of catching fish as compared to weaving cloth?
That is, which person has to give up weaving more cloth to catch a fish?

(e) Now imagine that a perfect perpetual summer has rolled around (a mini global warming). It's a balmy 78 degrees constantly and neither Bob nor Sarah have any use for clothing. Therefore, they don't have to weave. Instead, they only need fish and shelter to survive. One of these balmy days, Bob and Sarah meet for the first time on the top of the hill. The conversation turns to how well each can provide for themselves — how well each can fish and how well each can chop down trees. Despite their primitive early human brains, the two think that they can *together* produce more fish *and* more trees for shelter if each specializes in one activity. In order to maximize the total fish and total trees that they collectively produce, who should specialize in fishing? Who should specialize in tree chopping? Explain briefly why. **(4 points)**

(f) If Bob and Sarah each previously spent half of their 8 hour day fishing and half of their day chopping trees, how many fish and how many chopped trees could they produce? Now that they are specializing, how many more fish and how many more chopped trees can they produce? **(4 points)**

(g) Consider the following three quotations about specialization and markets. Indicate which of the following statements represent entirely *positive* statements and which have at least some *normative* component to them. (Note that some statements are paraphrases rather than direct quotations from the author.) **(3 points total, 1 point for each statement)**.

(1) *“A free market system allows all parties to compete, which ensures the best and most competitive project emerges, and ensures a fair, democratic process.”* – Sarah Palin

(2) *“I could probably earn more money in an hour of writing or teaching than I could in an entire week of cooking. But specialization breeds helplessness and dependence, and I don’t think people should be helpless or dependent, especially when it comes to food. People should know how to cook for themselves.”* – Michael Pollan, author of *The Omnivore’s Dilemma*

(3) *“Labor was the first price, the original purchase... It was not by gold or by silver, but by labor, that all wealth of the world was originally purchased.”* – Adam Smith, *The Wealth of Nations*

Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others. See the section on How Homework Will Work on the first page the problem set.